SPECULA

Six economies. Forty-four markets. Six rounds.

SPECULA, Latin for *watchtower* and the root of *speculation*, is a live macroeconomic trading simulation. Each team runs an investment fund. Six invented economies grow, slow, overheat and break on their own, and your fund trades their bonds, shares, currencies and commodities as they do. Every trade is entered on your own device, every price and score is calculated by the simulation, and nothing is judged.

The World

Aurelia *The Anchor*	**Calder** *The Workshop*
Verana *The Debtor*	**Kesh** *The Riser*
Norvik *The Well*	**Sarvan** *The Granary*

Six economies that trade with and react to one another. Each central bank sets rates by a rule shown on the desk, and each quarter's data grows out of the last: a team that understands the economics can see what is coming. Page 02 introduces each one.

Every Fund Starts Equal

₹1,000 Cr Fund value, all in cash	**3×** Most you may hold, against fund value	**100%** Risk limit, shown live on the desk	**0.10%** Cost of a trade, in calm markets
4 Quarters in every round	**2 min** Trading window each quarter	**24** Forecasts, one a quarter	**₹** Every price and value is in rupees

Positions, prices and fund values carry over from round to round: the six rounds are one continuous story.

How Every Quarter Works

01 The Data Drops
The quarter's figures for all six economies appear on every screen.

02 The Window Opens
A two-minute clock starts on every device and the desk unlocks.

03 Trade and Forecast
Buy, sell and enter the quarter's forecast. Change it as often as you like.

04 Stay Alert
A crisis may break at any moment. Prices react at once.

05 The Market Settles
At zero the desk locks, central banks act and prices reset.

06 Results
Fund values and standings appear on every screen at once.

The World

Six invented economies, each built on a different way a country can earn its living, and fail to.

Aurelia

The Anchor · crown

Large and wealthy, with the deepest bond market and the currency the world holds in reserve. Its banks lend across the globe, and when other markets panic, money runs here.

2.0%	4.2%	4.0%	90%
Growth	Inflation · target 4%	Policy rate	Debt to GDP

Calder

The Workshop · mark

The world's factory. It exports manufactured goods to everyone, saves heavily, runs a large trade surplus, and keeps the mark inside a band. It buys oil and copper in bulk.

5.5%	2.0%	3.0%	60%
Growth	Inflation · target 3%	Policy rate	Debt to GDP

Verana

The Debtor · florin

Old, wealthy and slow. Its government owes more than one and a half times its yearly output, prices barely rise, and its own banks hold most of its government's bonds.

0.8%	1.2%	1.0%	155%
Growth	Inflation · target 2%	Policy rate	Debt to GDP

Kesh

The Riser · dinar

A fast-growing emerging economy with high interest rates. It borrows heavily abroad in crowns and fixes the dinar at 50 to the crown, defending the peg with its reserves.

6.5%	7.0%	9.0%	55%
Growth	Inflation · target 5%	Policy rate	Debt to GDP

Norvik

The Well · krone

An oil exporter. Its budget, its trade and the krone rise and fall with the price of oil, and it saves the surplus of good years in a national wealth fund.

2.5%	3.0%	3.5%	30%
Growth	Inflation · target 3%	Policy rate	Debt to GDP

Sarvan

The Granary · ora

A frontier economy that sells wheat and copper to the world. Its ora floats freely, inflation runs high, and its markets are thin, so trading there costs more.

4.0%	9.0%	11.0%	70%
Growth	Inflation · target 6%	Policy rate	Debt to GDP

Figures are each economy's position at the start of Round 01, at annual rates.

How They Are Linked

No economy stands alone. Calder sells its goods to Aurelia and Verana, and buys oil from Norvik and copper and wheat from Sarvan. Kesh borrows from Aurelian lenders, and Verana borrows mostly from itself. Every central bank sets its rate by its own rule, shown on the desk, but all of them watch Aurelia's. A change in one economy travels to the others through trade, prices, interest rates and the movement of money.

The Markets

Forty-four instruments in eight families. Every one can be bought or sold short.

Government Bonds	14	**2-year and 10-year bonds of all six economies · Aurelia Inflation-Linked 10-year · Kesh Crown Bond 5-year** Prices move opposite to interest rates, and the 10-year moves furthest. The inflation-linked bond's value rises with Aurelian inflation, so it gains on an ordinary bond when prices surprise upward. The Crown Bond is Kesh borrowing in crowns: no dinar risk for you, but it falls as Kesh looks less able to pay.
Corporate Credit	02	**Aurelian Investment Grade · Aurelian High Yield** Baskets of company bonds. Investment grade lends to strong companies; high yield lends to weaker ones, pays far more interest and falls hardest when the economy turns.
Shares	08	**Market indices of all six economies · Aurelian Banks · Norvik Energy** Market indices follow growth, profits and the cost of borrowing. The two sector indices are narrower: bank shares answer to lending and bad loans, energy shares to the price of oil and gas.
Currencies	06	**Crown · mark · florin · dinar · krone · ora, each priced in rupees** Holding a currency earns that country's short-term rate. A currency rises when money flows into its country and falls when money leaves.
Rate Futures	06	**3-month rate futures for all six economies** A bet on next quarter's policy rate. Priced as 100 minus the rate the market expects, and settled at 100 minus the rate the central bank actually sets: buy if you expect rates lower than the market does, sell if higher.
Commodities	04	**Oil · Copper · Wheat · Gold** Oil follows world demand and anything that disturbs supply. Copper follows industry and building, wheat follows harvests and food supply, and gold pays nothing but is bought when investors are afraid.
Volatility	01	**The Fear Index** Measures how violently share prices are expected to move. It jumps when markets panic and drifts down when they are calm, and holding it costs a fee every quarter.
Default Protection	03	**Kesh Government · Verana Government · Aurelian Banks** Insurance against a borrower failing to pay. It costs a premium every quarter, its price rises as default looks more likely, and it pays out in full if the default happens.

Every price on the desk is in rupees. The desk shows each instrument's price, its change since the last quarter, and what holding it earns or costs.

The Trading Desk

One fund and forty-four markets: how a trade works, and what holding a position earns and costs.

Placing a Trade

Buy	Sell Short	Close
Hold an instrument, in ₹ Cr. You gain when its price rises.	Sell an instrument you do not own. You gain when its price falls.	End a position, all of it or part, and take the profit or loss.

All positions together, buys and shorts both counted, may reach three times your fund value; the desk refuses any order beyond that. Unused cash earns the rupee cash rate shown on the desk; positions beyond your cash are funded by borrowing at that rate plus 1% a year.

Earning and Paying While You Hold

You Earn	Interest on government and company bonds, and the short-term rate of any currency you hold, paid every quarter.
You Pay	The premium on default protection, the fee on the Fear Index, and the interest or rate on anything you have sold short.
Settled Each Quarter	Rate futures settle at the end of every quarter and roll into the next quarter's contract. Shares, commodities and currencies pay nothing beyond their price change, except the rate a currency earns.

The Price You Pay

Every trade costs a spread: 0.10% in calm markets, more under strain, always shown on the desk. Prices also answer to the room. When many funds buy the same instrument in one quarter, its price is pushed up before it settles, and heavy selling pushes it down. One fund moves a market a little; every fund doing the same thing moves it a lot.

Your Fund Screen

Beside the markets, the desk shows your cash, every open position with its profit or loss, your total exposure against the 3× limit, your Risk Used, and your fund value. It updates as you trade and again when the quarter settles.

Working as a Team

Members may sign in on their own devices; all see the same desk, and any member's trade is the fund's trade. Decide before the round who places orders and who reads the data, so no trade is placed twice.

Risk, Forecasts and Crises

How much you may risk, what you must predict, and what may happen without warning.

The Risk Limit

Risk Used Live, on the desk	An estimate of what your fund could lose in a bad quarter, shown as a share of the limit. It grows with the size of each position and with how sharply that instrument has been moving. Positions that offset one another count for less.
Forced Liquidation Above 100% at settlement	If Risk Used is above 100% when the quarter settles, the system closes your largest positions until it falls to 80%, at a price 2% worse than the market, and records a forced liquidation against your fund.
Risk Moves Too Even if you do nothing	When markets turn violent, the same positions carry more risk. Risk Used can rise past the limit without a single new trade. Watch it until the bell.

Forecasts

Every quarter the desk names one figure to forecast for the next release: a country's inflation, growth, unemployment or policy rate.

Scoring

Up to 10 points a forecast. You lose 1 point for every 0.1 percentage point you are off.

Over the Event

Twenty-four forecasts in all, worth up to 240 points. A blank forecast scores zero.

Crises

At any moment the Event Head may release a crisis. Headlines and posts appear on every screen and prices react at once. A crisis may also change the data, the rules or the cost of trading for the rest of the round. Nothing about a crisis is announced in advance, and the desk stays open through it for as long as the window runs.

Reading the Data

Each release gives growth, inflation, unemployment, the policy rate, the budget balance and the current account for every economy, with figures particular to each, such as reserves or oil output. Each central bank's rule for setting rates is on the desk. Everything the simulation does follows from the data and the rules you can see. A calculator is allowed, and you will want one.

THE SIX ROUNDS

01
MARKETS IN NORMAL TIMES
Four quarters · about 10 minutes

02
INFLATION AND INTEREST RATES
Four quarters · about 10 minutes

03
COMMODITIES AND TRADE
Four quarters · about 10 minutes

04
EMERGING MARKETS AND CURRENCIES
Four quarters · about 10 minutes

05
BANKING AND CREDIT
Four quarters · about 10 minutes

06
SOVEREIGN DEBT AND RECOVERY
Four quarters · the final round

The economy runs on from round to round, and what each round brings is revealed only when it opens. Some rounds add a decision to the desk; it appears on screen, with its rules, when that round begins.

HOW FUNDS ARE RANKED

Funds are ranked by the **SPECULA Score**, out of 100, calculated after the last quarter of Round 06. It rewards a fund that read the economy well over one that gambled and got lucky.

60 — **RISK-ADJUSTED RETURN**
Your fund's total return divided by how much its value swung from quarter to quarter, less a penalty for its deepest fall from a peak. The best fund in the room scores 60, the rest in proportion; a fund that ends below ₹1,000 Cr scores zero here.

25 — **FORECAST ACCURACY**
Your forecast points from all 24 quarters, scaled so that a perfect record scores 25.

15 — **RISK DISCIPLINE**
Every fund starts with 15. Each forced liquidation costs 5, down to zero.

After every quarter your screen shows your fund value, your profit and loss on each position, your Risk Used and your standing. If two funds tie on the SPECULA Score, the higher final fund value ranks first.

RULES OF PLAY

01 Each team runs one fund. Members may sign in to it on their own devices.

02 Trades and forecasts are accepted only while the window is open. Late orders are refused.

03 Positions stay open from quarter to quarter, and round to round, until you close them.

04 Every price, result and score is calculated for all funds at the same moment. There is no judging and no appeal.

05 Fund codes are private. Speaking to other teams or looking at their screens is not permitted.

06 Calculators are allowed. No device other than your desk and calculator may be used.